

# A Social Computing Analysis of Flood Control Discussions in the Philippines

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**Abstract**—Following recurring flood events in the Philippines, public discourse has shifted from disaster response toward debates on political accountability. This study analyzes 6,830 public comments and articles collected from Rappler and YouTube between July 2024 and November 2025. Using Latent Dirichlet Allocation (LDA) for topic modeling, XLM-RoBERTa for sentiment analysis, and Social Network Analysis (SNA), we identify two dominant meta-narratives: (1) systemic corruption and (2) technical infrastructure failure. Results reveal a stark platform divergence: YouTube comments are approximately 73% negative, driven by moralized blame (e.g., "pera ng bayan"), while Rappler remains comparatively neutral. Network analysis identifies "collective blame" hubs linking flood control directly to specific political actors. These findings highlight how digital platforms shape crisis perception, suggesting that unmoderated digital engagement accelerates public cynicism. This study demonstrates how computational social science can quantify platform-specific distortions in crisis discourse, offering empirical insight into how digital architectures shape public perceptions of national crises.

## Index Terms

Social computing, topic modeling, sentiment analysis, flood control, Philippines.

## I. Introduction

The Philippines is frequently cited as one of the most disaster-prone countries in the world, but recent discourse regarding flood control has shifted from resilience toward accountability. In November 2025, reports of substandard flood defense projects sparked mass protests and significant online backlash. The controversy centered on allegations that billions of pesos allocated for flood control had been misused or redirected, leading to widespread public anger. For many citizens, social media platforms have become the primary “digital public square” for expressing these frustrations.

However, the way this discourse unfolds varies significantly across platforms. Prior research shows that while news media provides structured crisis framing, user-generated comment sections often evolve into “unruly” spaces marked by hostility, polarization, and emotional escalation. We ground this analysis in ‘platform ecology,’ positing that the distinct technical architectures of YouTube (algorithmic recommendation) and Rappler (editorial curation) influence how the flood crisis is discussed. For policymakers, understanding this divergence is crucial in distinguishing between platform-specific noise and genuine public sentiment.

In this study, we use a social computing framework to compare the rhetorical storylines and behavioral patterns present in Rappler articles versus YouTube comments during the 2024–2025 flood control controversy. We address the following research questions:

1. What are the dominant rhetorical storylines in the data?
2. How does sentiment differ between professional (news) and public (comment) discourse?
3. What behavioral dynamics drive engagement in these networks?

Computational analysis allows us to systematically compare these two discursive spaces and quantify how public frustration manifests online. This paper contributes: (1) a comparative analysis of flood discourse across professional (news) and public (social media) channels; (2) a mixed-method pipeline combining LDA topic modeling, transformer-based sentiment analysis, and network analysis to quantify the “affective divergence” between platforms; and (3) empirical evidence of “collective blame” networks that link infrastructure failure directly to political corruption. Finally, this study argues for the necessity of mixed methods. While computational techniques allow us to scale the analysis to thousands of comments, qualitative reading remains essential to decode specific cultural narratives like ‘pera ng bayan’ (public money).

## II. Related Literature

- A. Social Computing in Disaster Management Social media is no longer just a communication tool; it is a mechanism for real-time crowdsourcing during disasters. In the Philippines, platforms like Facebook, YouTube, and X (Twitter) play a major role in shaping how the public perceives government efficacy. However, this ecosystem is often complicated by misinformation and

emotionally charged narratives that can drown out official communications.

- B. Platform Ecology and Discourse Scholars have noted a "hostility gap" between journalistic content and user comments. While news articles generally strive for neutrality, comment sections, particularly on video platforms, tend to amplify high-arousal emotions like anger. Research by Brady et al. [5] further suggests that moral-emotional words significantly increase diffusion in social networks, creating amplification loops that prioritize negativity over factual discourse. This study builds on Montiel et al.'s [1] framework of crisis meaning-making by quantifying the specific "affective divergence" between Rappler's professional discourse and the public discourse found on YouTube.

### III. Methodology

We adapted the computational pipeline used by Alharbi et al. [2], integrating data collection, text preprocessing, and multi-method analysis (topic modeling, sentiment classification, and network analysis). Although their work centers on Arabic offensive language detection, we adapt their preprocessing and transformer-based classification approach for our multilingual sentiment task.

- A. Dataset Description The dataset consists of 6,830 documents collected between July 27, 2024, and November 12, 2025. It is stratified into two distinct categories of discourse: professional news reporting and analysis, comprising 37 articles from Rappler, and public engagement, comprising 6,793 user comments from YouTube videos. Prior to analysis, the data underwent rigorous validation to remove duplicate entries and corrupted text files.
- B. Preprocessing and Text Cleaning To ensure semantic clarity, we applied a three-tier stopword removal system: (1) basic English stopwords (e.g., "the", "is"), (2) common Tagalog stopwords (e.g., "ang", "mga", "wala"), and (3) domain-specific noise tokens (e.g., "rappler", "youtube", "2025"). Finally, all text was lowercased, tokenized, and stripped of non-semantic artifacts such as URLs and emojis. The complete preprocessing pipeline, including the custom stopword filtration, is documented in the accompanying notebook 02\_Preprocessing\_and\_EDA\_Flood.ipynb.
- C. Computational Models We applied three primary computational techniques to analyze the corpus. First, we used Latent Dirichlet Allocation (LDA) [3] for topic modeling. The value  $k=7$  was selected after comparing coherence scores across  $k=5$  to  $k=12$ , where  $k=7$  provided the optimal balance of interpretability and topic separation. Second, we

employed the XLM-RoBERTa transformer model [4] for sentiment classification, selected specifically for its robust performance on code-switched (Taglish) text. We used the multilingual xlm-roberta-base checkpoint, fine-tuned for three-way sentiment classification (positive, neutral, negative), assigning labels based on the highest-probability class. Third, we conducted Social Network Analysis (SNA) by constructing a bigram co-occurrence graph of the top 75 most frequent word pairs to identify semantic hubs and "collective blame" clusters.

## IV. Results and Discussion

### A. Platform Demographics

Table I summarizes the distribution of data across the two platforms. The corpus is heavily skewed toward YouTube comments, reflecting the high volume of public engagement on social media compared to curated news articles.

**Table I. Dataset Distribution**

| Platform     | Count        | Percentage  |
|--------------|--------------|-------------|
| YouTube      | 6,793        | 99.5%       |
| Rappler      | 37           | 0.5%        |
| <b>Total</b> | <b>6,830</b> | <b>100%</b> |

Note: The significant disparity in sample size ( $N=37$  vs.  $N=6,793$ ) is inherent to the platform comparison, as professional journalistic output is far lower in volume than mass public commentary.

### B. Dominant Storylines

Topic labeling follows the narrative approach of Montiel et al. [1]. As detailed in the methodology,  $k=7$  was selected as it offered the optimal balance of topic distinctiveness and semantic coherence. Our LDA analysis highlights two contrasting meta-narratives in the discussion:

- 1) **Accountability Narratives:** The most prominent storylines focus on governance. In Topic 0 (see Fig. 1), top terms like *marcos*, *commission*, and *probe* show that citizens view the floods as a failure of oversight that requires investigation. This links closely with Topic 6 (see Fig. 3), where terms like *corruption* and *government* suggest that users see corruption not as a one-time event, but as the root cause of infrastructure failure. We also observed Topic 1 (see Fig. 2) and Topic 4 (see Fig. 4), which directed

specific scrutiny toward local officials and high-level actors, respectively.

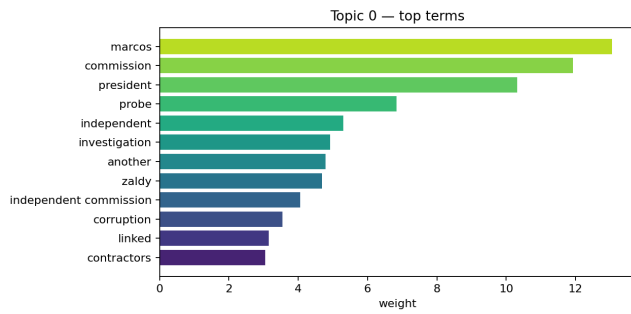


Fig. 1. Top terms for Topic 0 (Accountability & Investigation)

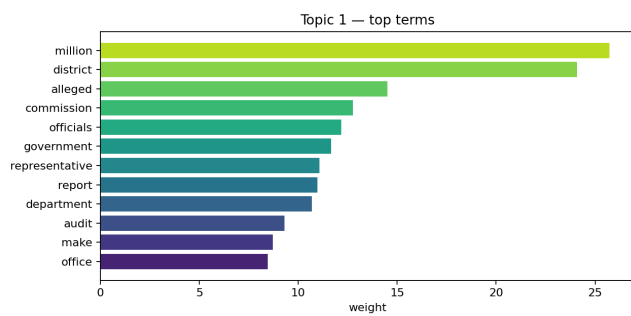


Fig. 2. Top terms for Topic 1 (Audit & Local Officials)

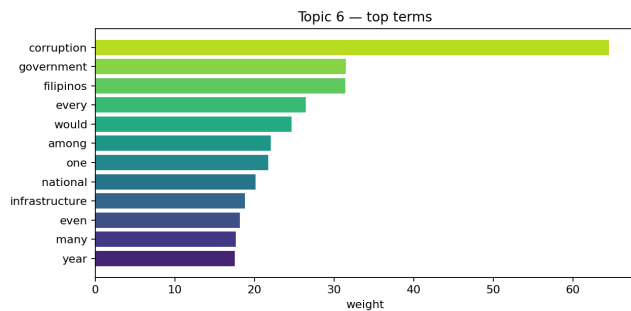


Fig. 3. Top terms for Topic 6 (Structural Corruption)

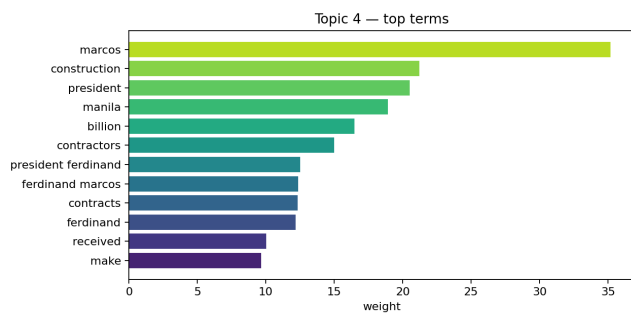


Fig. 4. Top terms for Topic 4 (High-level Actors)

2) **Material & Technical Narratives:** Technical concerns appeared in Topic 2 (see Fig. 5), Topic 3 (see Fig. 6), and Topic 5 (see Fig. 7). However, even when discussing technical terms like budget, works, or construction, the language was frequently paired with blame. This creates a hybrid narrative of "technical incompetence caused by corruption," where engineering failures are interpreted as evidence of stolen funds. Collectively, the dominance of these narratives suggests that the public interprets the flood primarily as a failure of governance rather than a natural disaster.

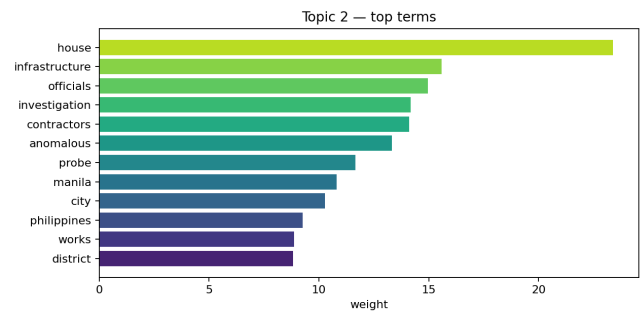


Fig. 5. Top terms for Topic 2 (Infrastructure)

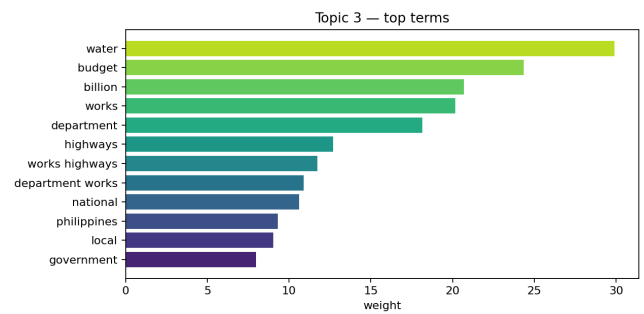


Fig. 6. Top terms for Topic 3 (Public Funds)

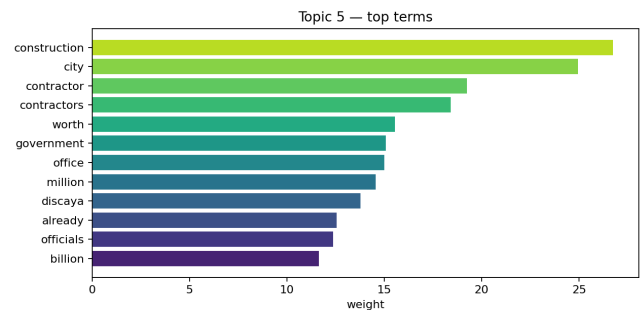


Fig. 7. Top terms for Topic 5 (Construction Projects)

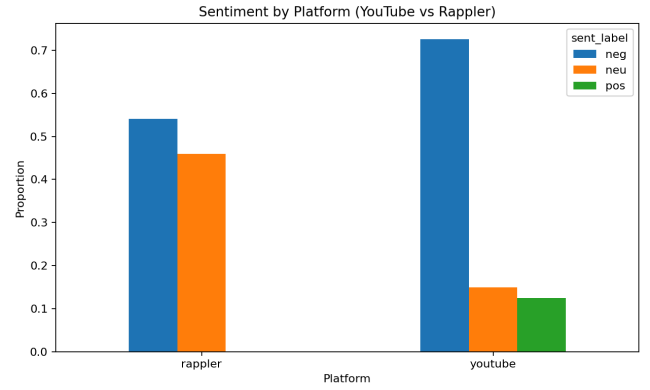
Collectively, these seven topics converge into two meta-narratives: one focusing on institutional corruption (Topics 0, 1, 4, 6) and another on technical failure (Topics 2, 3, 5). The dominance of the corruption narrative suggests that the public interprets the flood primarily as a failure of governance rather than nature.

#### C. Sentiment Analysis: Cross-Platform Sentiment Patterns

We observed a significant disparity in sentiment between platforms (see Fig. 8). YouTube functions as a high-arousal emotional environment, with nearly three-quarters of all comments classified as negative. In contrast, Rappler articles and comments maintain a more balanced, neutral tone typical of journalistic discourse. Notably, positive sentiment was relatively limited (12.4%) and often limited to sarcasm or unrelated praise.

**Table II. Sentiment by Platform**

| Platform | Negative         | Neutral          | Positive       | Total |
|----------|------------------|------------------|----------------|-------|
| Rappler  | 20<br>(54.1%)    | 17<br>(45.9%)    | 0 (0.0%)       | 37    |
| YouTube  | 4,932<br>(72.6%) | 1,021<br>(15.0%) | 840<br>(12.4%) | 6,793 |



**Fig. 8. Sentiment distribution differences between Rappler (News) and YouTube (Public Comments)**

Topics related to corruption (Topics 0, 4, and 6) showed the highest density of negative sentiment. Cross-referencing with our topic model, Topic 6 (Structural Corruption) exhibited the highest negative density, confirming the link between systemic critique and anger. Even infrastructure-focused topics contained mixed sentiment, confirming that in this context, infrastructure is a deeply political subject rather than a purely operational one.

#### D. Behavioral Patterns & Engagement

The platform-level negativity observed in the sentiment analysis provides the emotional backdrop that explains the engagement patterns identified in the network analysis.

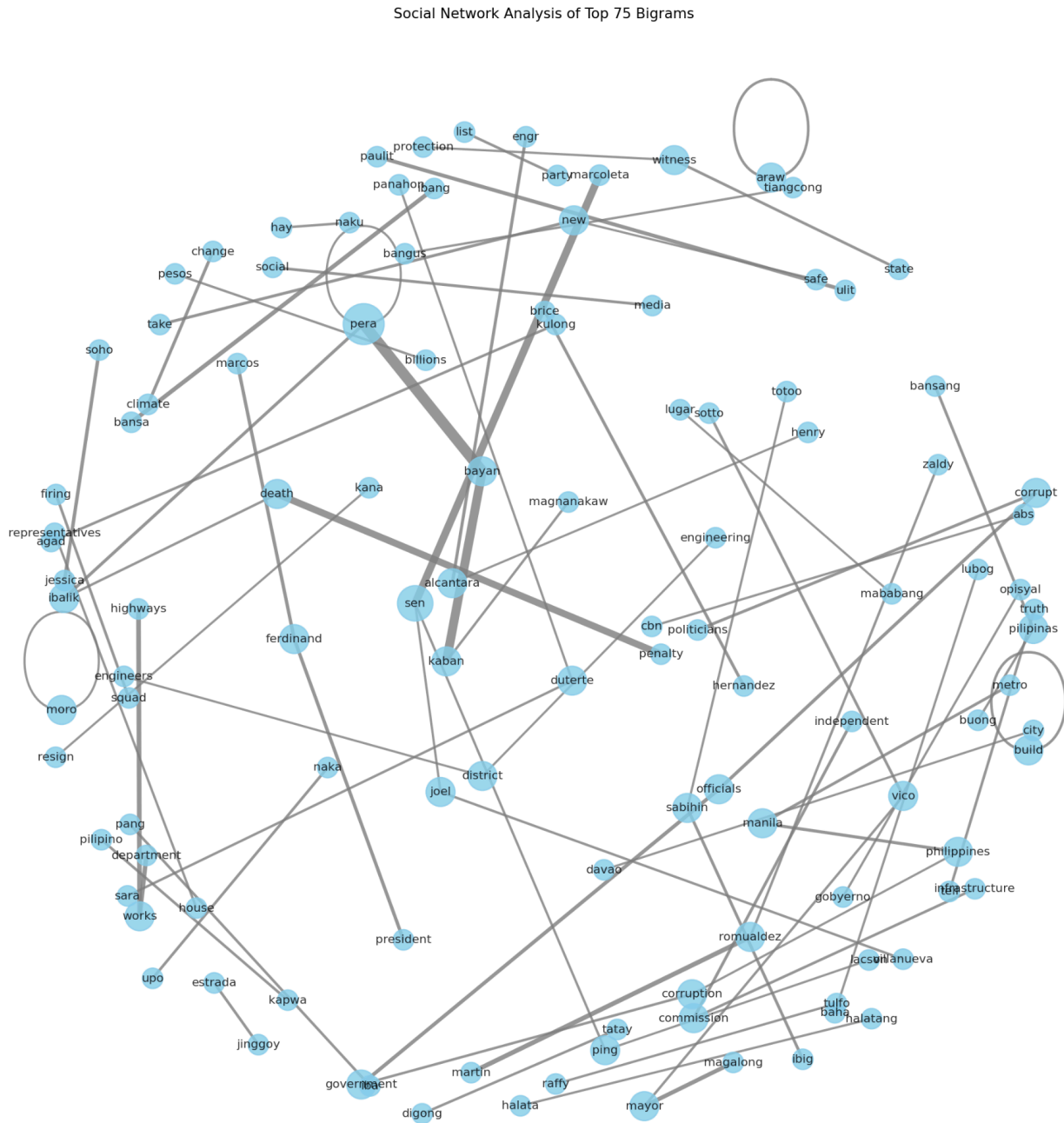


Fig. 9. Bigram co-occurrence network showing "collective blame" hubs linking political actors to fiscal terms

- 1) **Network Structure (Collective Blame):** Our Social Network Analysis (SNA) graph (see Fig. 9) shows that political actors and financial terms (pera, bayan, commission) act as the central "hubs" of the discussion. These terms form a cohesive "blame network," suggesting that the community is tightly connected by shared grievances rather than divided into separate factions. The centrality of the 'pera-bayan' hub correlates directly with the content of Topic 3, bridging the structural topic modeling results with behavioral co-occurrence patterns. Edge thickness in Fig. 9 reflects the frequency of bigram co-occurrence, highlighting which conceptual links dominate the conversation.
- 2) **The Visibility Economy of Outrage:** When analyzing engagement, we found that outrage drives visibility. Fig. 10 displays the top 10 most-liked comments, which were exclusively negative. The single most-liked comment, "Wala na yan 26 billion pesos..." ("That 26 billion pesos is gone..."), demonstrates that narratives regarding stolen public wealth receive the most algorithmic amplification, effectively drowning out neutral or technical discussions. The popularity of these comments confirms that the platform rewards emotionally charged, blame-oriented narratives.

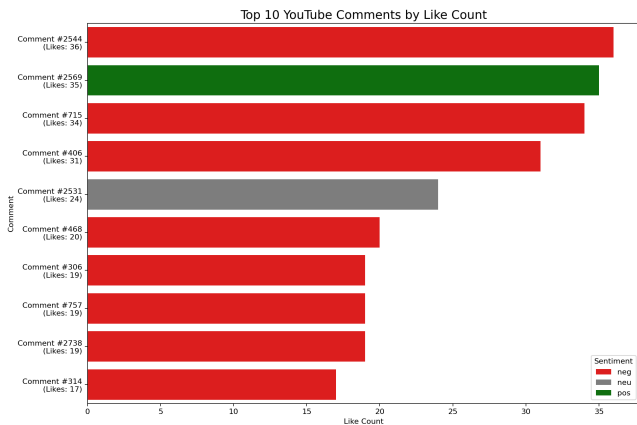


Fig. 10. The top 10 most-liked comments on YouTube, all showing negative sentiment.

## V. Limitations

While this study provides significant insights into digital discourse, several limitations should be noted.

First, the dataset introduces sampling bias. YouTube comments represent a self-selected demographic of highly active users and may not reflect the sentiments of the broader Filipino population. Furthermore, comparing journalistic articles (Rappler) with user-generated comments (YouTube) creates a genre mismatch; the neutrality of Rappler may stem from professional editorial standards rather than organic public sentiment. Comparing long-form journalism with short informal comments introduces a structural bias; news articles prioritize neutrality by design, whereas comment sections are architected to maximize engagement through emotional arousal.

Second, while XLM-RoBERTa is optimized for multilingual text, it faces challenges with the linguistic nuances of Philippine English and Tagalog code-switching. Specifically, the model may struggle to detect sarcasm and irony, which are prevalent in Filipino political discourse (e.g., using "magaling" or "best president" in a derogatory context), potentially skewing positive sentiment scores.

Third, the LDA topic modeling approach relies on bag-of-words assumptions, which can obscure complex narrative structures. Similarly, the Social Network Analysis (SNA) relies on bigram co-occurrences; while this identifies semantic hubs, it does not fully capture the causal or argumentative links between concepts. Furthermore, this study provides a static snapshot of the dataset and does not account for temporal dynamics, such as how public sentiment evolved immediately following specific news cycles or government announcements.

Lastly, data collection was restricted to publicly accessible endpoints in strict compliance with platform terms of service to ensure user privacy.

## VI. Conclusion

Ultimately, this study suggests that online discourse regarding flood control in the Philippines is driven less by engineering concerns and more by a breakdown in political trust. While Rappler frames the crisis through institutional processes, YouTube users frame it as a moral failure. With 73% negative sentiment and a network structure centered on blame, the public interprets these flood events not as natural disasters, but as political betrayals. Policymakers must recognize that flood control is no longer just an engineering challenge but a communication crisis. Effective governance requires transparent digital strategies to counter the algorithmic amplification of distrust.

## VII. Future Work

To build upon these findings, future research should expand the scope of analysis in three directions. First, given the static nature of this dataset, a temporal analysis could be conducted to map how public sentiment

evolves in real-time during the onset of a typhoon versus the post-disaster recovery phase. Second, as younger demographics increasingly rely on visual media, this framework should be applied to short-form video platforms (e.g., TikTok) to understand how audiovisual memes and algorithmic trends accelerate narratives of accountability. Finally, future iterations could employ fine-tuned transformer models specifically trained on Philippine political satire to better detect irony and sarcasm, which remain challenging for general multilingual models.

## VIII. References

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